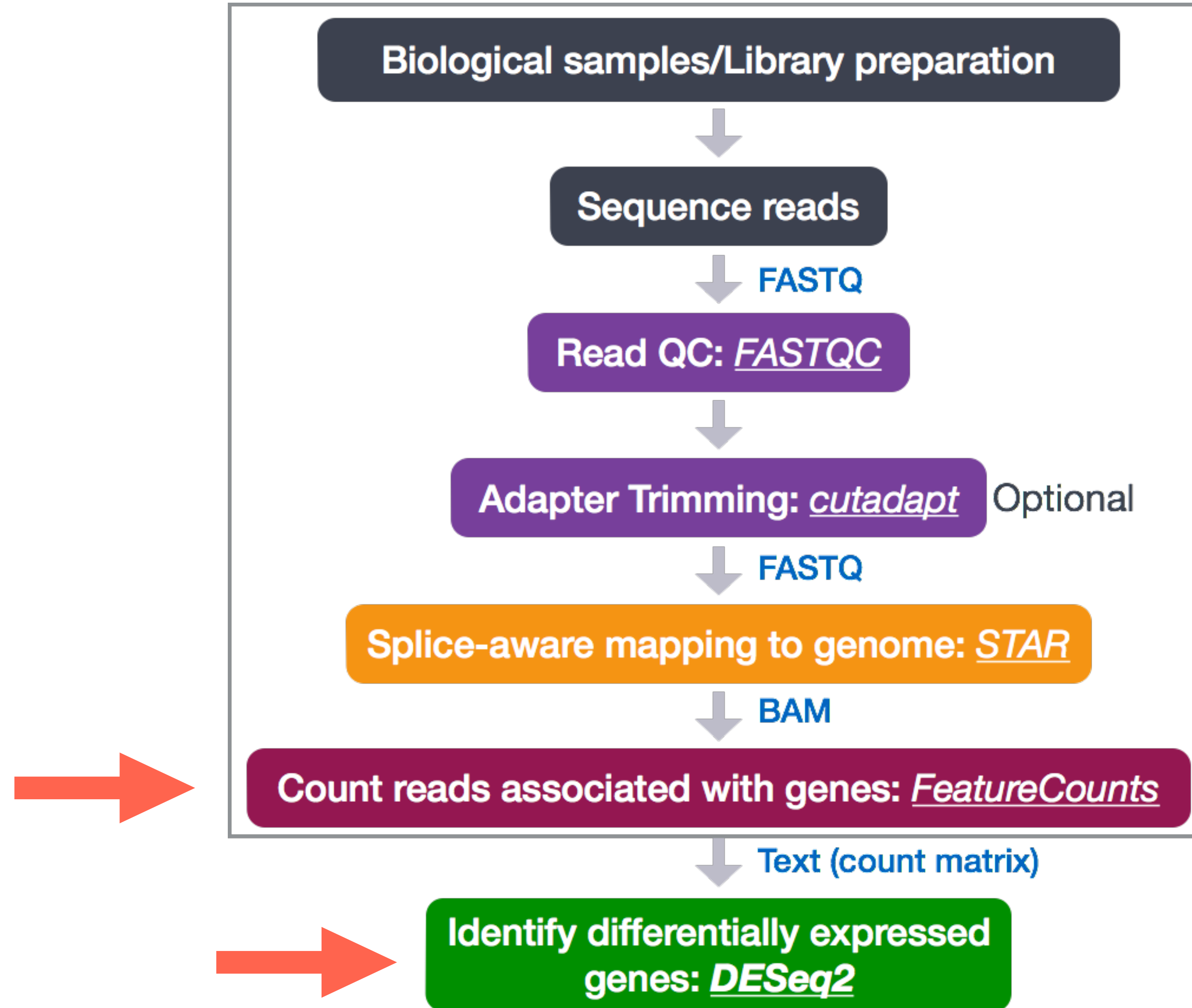


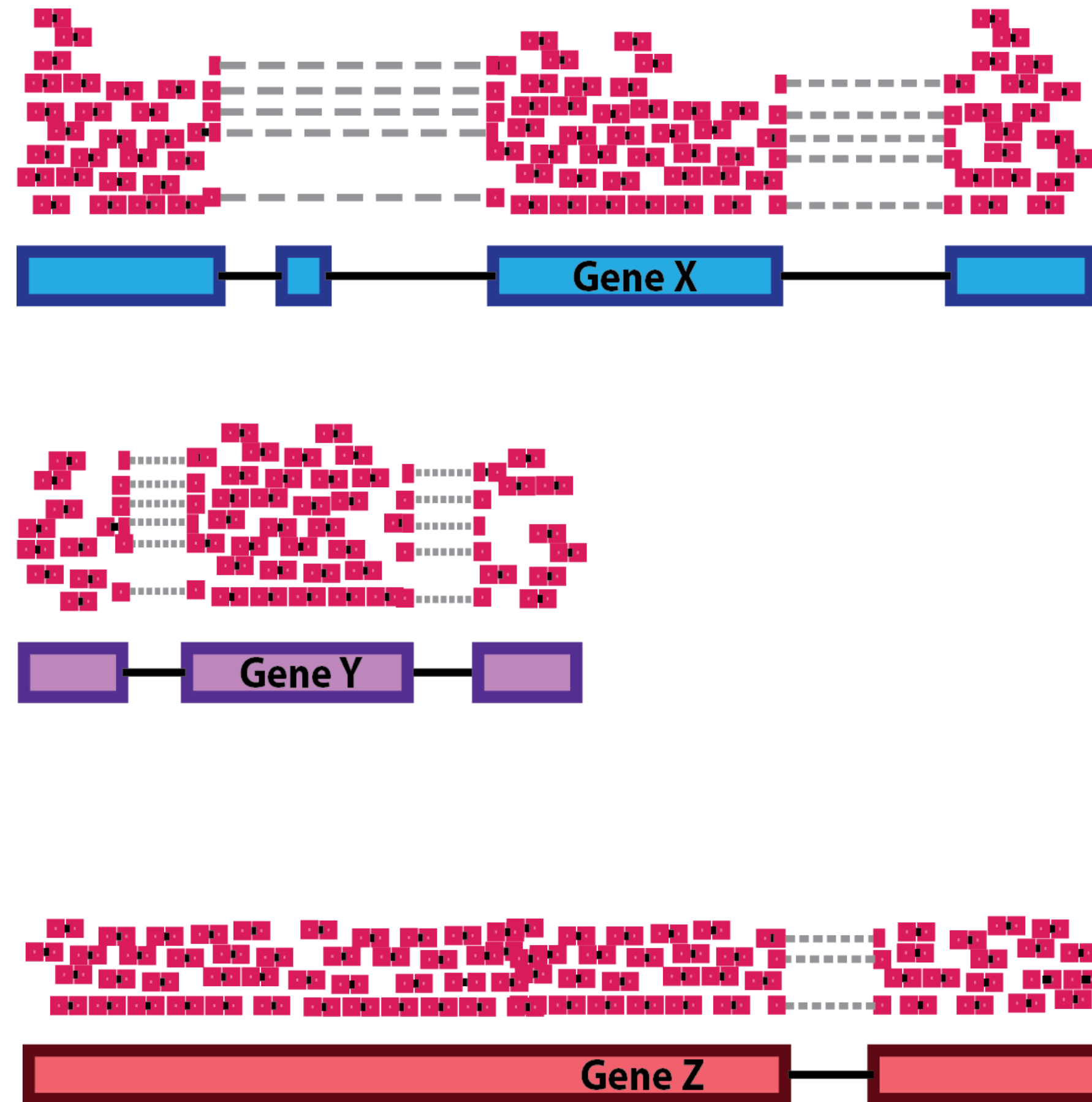
RNA-seq analysis workflow



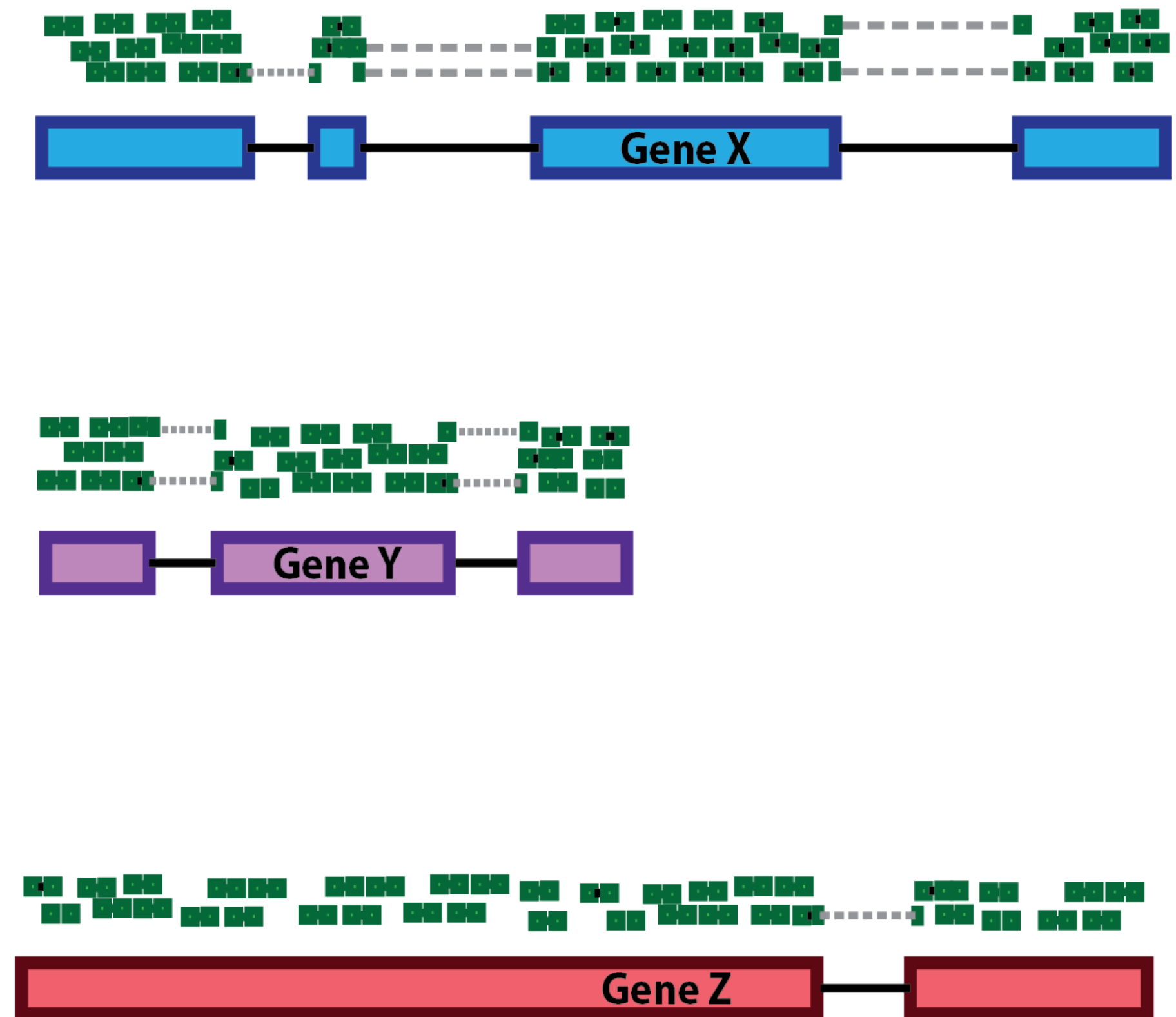
We will use EdgeR!

RNA-seq sample normalization strategies

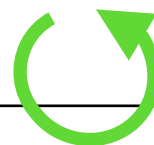
Sample A Reads



Sample B Reads



RNA-seq sample normalization strategies



	Sample A	Sample B
Gene X	50	10
Gene Y	36	6
Gene Z	42	55



$$\text{RPM or CPM} = \frac{\text{Number of reads mapped to gene} \times 10^6}{\text{Total number of mapped reads}}$$

$$\text{RPKM} = \frac{\text{Number of reads mapped to gene} \times 10^3 \times 10^6}{\text{Total number of mapped reads} \times \text{gene length in bp}}$$

Normalized count values output by RPKM/FPKM methods are not comparable between samples!
Total number of RPK/FPKM normalized counts for each sample will be different.

Importantly, these approaches do NOT account for compositional differences between libraries.
Think of library compositional differences as different samples/treatments containing different active genes.

How can we account for sequencing depth and RNA composition when normalizing data?

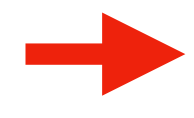
DESeq2's median of ratios	counts divided by sample-specific size factors determined by median ratio of gene counts relative to geometric mean per gene	sequencing depth and RNA composition	gene count comparisons between samples and for DE analysis ; NOT for within sample comparisons
EdgeR's trimmed mean of M values (TMM)	uses a weighted trimmed mean of the log expression ratios between samples	sequencing depth, RNA composition, and gene length	gene count comparisons between and within samples and for DE analysis

How can we account for sequencing depth and RNA composition when normalizing data?

An example using the Trimmed Mean of M-values (TMM) normalization with EdgeR.

The following presentation is modeled after Joshua Starmer's presentation on the topic (statquest.org)

edgeR - Trimmed Mean of M-values Normalization



	Sample #1	Sample #2	Sample #3
Gene 1	0	10	4
Gene 2	2	6	12
Gene 3	33	55	200
Gene 4	12	40	33
Gene5	0	0	0

Step 1 - Filter out all genes that are not transcribed

	Sample #1	Sample #2	Sample #3
Gene 1	0	10	4
Gene 2	2	6	12
Gene 3	33	55	200
Gene 4	12	40	33

edgeR - Trimmed Mean of M-values Normalization

Step 2 - Pick the reference sample.

One sample is chosen to be the reference sample and used to normalize all the other samples.
EdgeR tries to identify the most “average” sample to avoid scaling based on noisy samples.

	Sample #1	Sample #2	Sample #3
Gene 1	0	10	4
Gene 2	2	6	12
Gene 3	33	55	200
Gene 4	12	40	33

Scale each sample
by its read counts

Total Reads	47	111	249
-------------	----	-----	-----

	Sample #1	Sample #2	Sample #3
Gene 1	0/47	10/111	4/249
Gene 2	2/47	6/111	12/249
Gene 3	33/47	55/111	200/249
Gene 4	12/47	40/111	33/249

edgeR - Trimmed Mean of M-values Normalization

Step 2 - Pick the reference sample.

One sample is chosen to be the reference sample and used to normalize all the other samples.
EdgeR tries to identify the most “average” sample to avoid scaling based on noisy samples.

	Sample #1	Sample #2	Sample #3
Gene 1	0	10	4
Gene 2	2	6	12
Gene 3	33	55	200
Gene 4	12	40	33

Scale each sample
by its read counts

Total Reads	47	111	249
-------------	----	-----	-----

	Sample #1	Sample #2	Sample #3
Gene 1	0	0.09	0.02
Gene 2	0.04	0.05	0.05
Gene 3	0.70	0.50	0.80
Gene 4	0.26	0.36	0.13

Determine the 75th
quantile for each
sample

75th Quant.	0.26	0.36	0.13
-------------	------	------	------

75th quantile - 3/4 values (75%)
are less than or equal to 0.13

edgeR - Trimmed Mean of M-values Normalization

Step 2 - Pick the reference sample.

One sample is chosen to be the reference sample and used to normalize all the other samples.
EdgeR tries to identify the most “average” sample to avoid scaling based on noisy samples.

Determine the 75th
quantile for each
sample

	Sample #1	Sample #2	Sample #3
Gene 1	0	0.09	0.02
Gene 2	0.04	0.05	0.05
Gene 3	0.70	0.50	0.80
Gene 4	0.26	0.36	0.13

Average the
75th quantiles

75th Quant. $\frac{0.26 + 0.36 + 0.13}{3} = 0.25$

The reference sample is the sample with the 75th quantile closest to the average

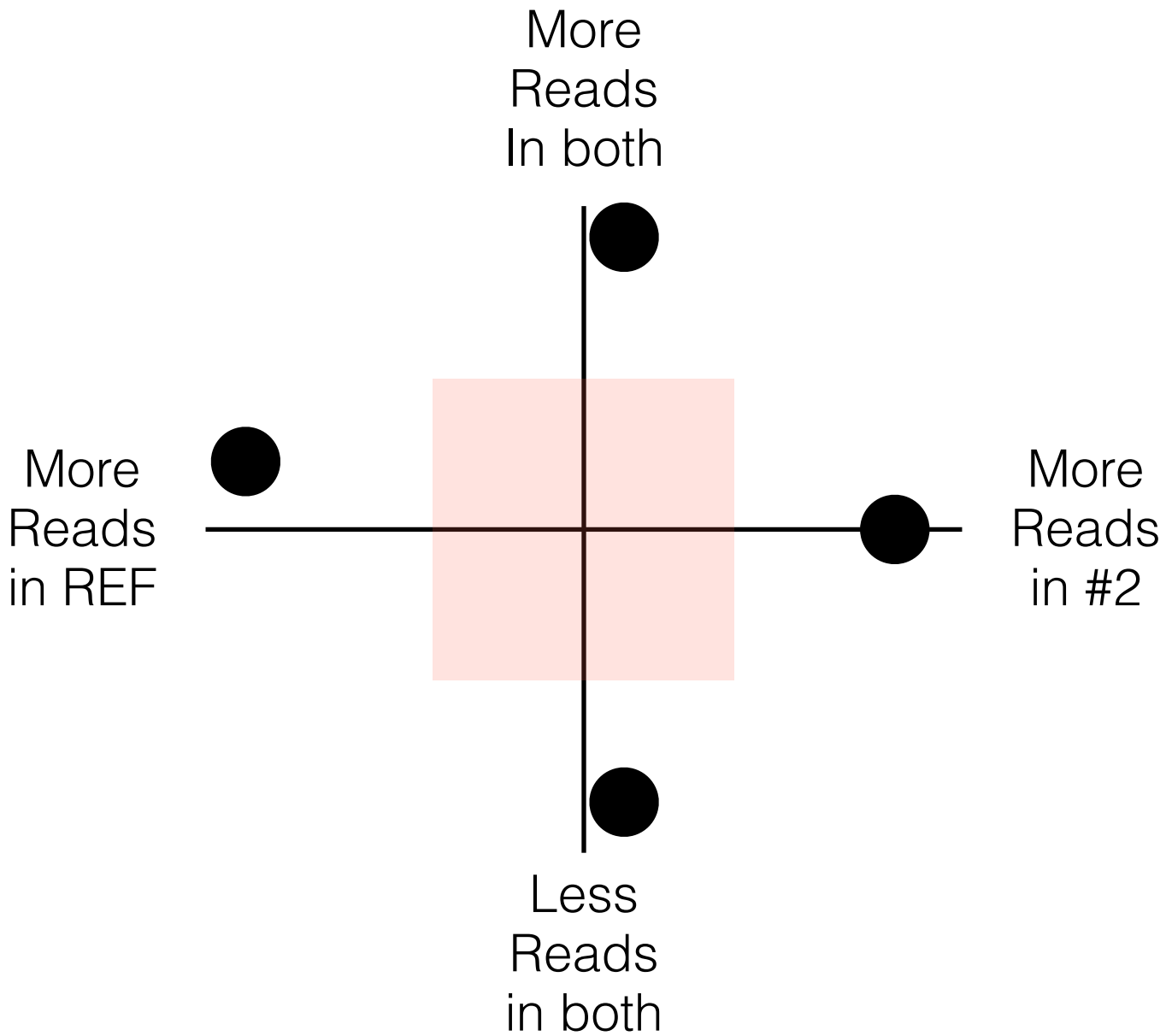
edgeR - Trimmed Mean of M-values Normalization

Step 3 - Select genes for calculating the scaling factors

This is done individually for each sample (edgeR's approach) and relative to the reference sample.

	Sample #1	Sample #2	Sample #3
Gene 1	0	10	4
Gene 2	2	6	12
Gene 3	33	55	200
Gene 4	12	40	33

What type of genes can we choose from?



EdgeR will exclude biased genes and pick genes from the middle for calculating scaling factors

edgeR - Trimmed Mean of M-values Normalization

Step 3 - Select genes for calculating the scaling factors

This is done individually for each sample (edgeR's approach) and relative to the reference sample.

	Sample #1	Sample #2
Gene 1	0	0.09
Gene 2	0.04	0.05
Gene 3	0.70	0.50
Gene 4	0.26	0.36
...	...	...
Gene X	0.13	0.15



	$\log_2(\text{Reference} / \text{Sample \#2})$
Gene 1	$\log_2(0/0.09) = -\text{Inf}$
Gene 2	$\log_2(0.04/0.05) = -0.32$
Gene 3	$\log_2(0.70/0.50) = 0.49$
Gene 4	$\log_2(0.26/0.36) = -0.47$
...	...
Gene X	$\log_2(0.13/0.15) = -0.21$

To identify biased genes

	Geometric Mean
Gene 1	$-\text{Inf}$
Gene 2	-4.48
Gene 3	-0.76
Gene 4	-1.71
...	...
Gene X	-2.84

genes that are highly and lowly transcribed
in both samples

edgeR - Trimmed Mean of M-values Normalization

Step 3 - Select genes for calculating the scaling factors

This is done individually for each sample (edgeR's approach) and relative to the reference sample.

	Sample #1	Sample #2
Gene 1	0	0.09
Gene 2	0.04	0.05
Gene 3	0.70	0.50
Gene 4	0.26	0.36
...	...	...
Gene X	0.13	0.15



	$\log_2(\text{Reference} / \text{Sample \#2})$
Gene 2	$\log_2(0.04/0.05) = -0.32$
Gene 3	$\log_2(0.70/0.50) = 0.49$
Gene 4	$\log_2(0.26/0.36) = -0.47$
...	...
Gene X	$\log_2(0.13/0.15) = -0.21$

	Geometric Mean
Gene 2	-4.48
Gene 3	-0.76
Gene 4	-1.71
...	...
Gene X	-2.84

edgeR - Trimmed Mean of M-values Normalization

Step 3 - Select genes for calculating the scaling factors

This is done individually for each sample (edgeR's approach) and relative to the reference sample.

	Sample #1	Sample #2
Gene 1	0	0.09
Gene 2	0.04	0.05
Gene 3	0.70	0.50
Gene 4	0.26	0.36
...	...	...
Gene X	0.13	0.15



Sort both tables from low to high

	$\log_2(\text{Reference} / \text{Sample \#2})$
Gene 4	-0.47
Gene 2	-0.32
Gene N	-0.21
...	...
Gene 3	0.49

	Geometric Mean
Gene 2	-4.48
Gene N	-2.84
Gene 4	-1.71
...	...
Gene 3	-0.76

edgeR - Trimmed Mean of M-values Normalization

Step 3 - Select genes for calculating the scaling factors

This is done individually for each sample (edgeR's approach) and relative to the reference sample.

	Sample #1	Sample #2
Gene 1	0	0.09
Gene 2	0.04	0.05
Gene 3	0.70	0.50
Gene 4	0.26	0.36
...	...	...
Gene X	0.13	0.15

Filter out the top 30% and the bottom 30% biased genes



Genes that remain in both lists are used for calculating scaling factors



Filter out the top 5% and the bottom 5% of the highly and lowly transcribed genes

edgeR - Trimmed Mean of M-values Normalization

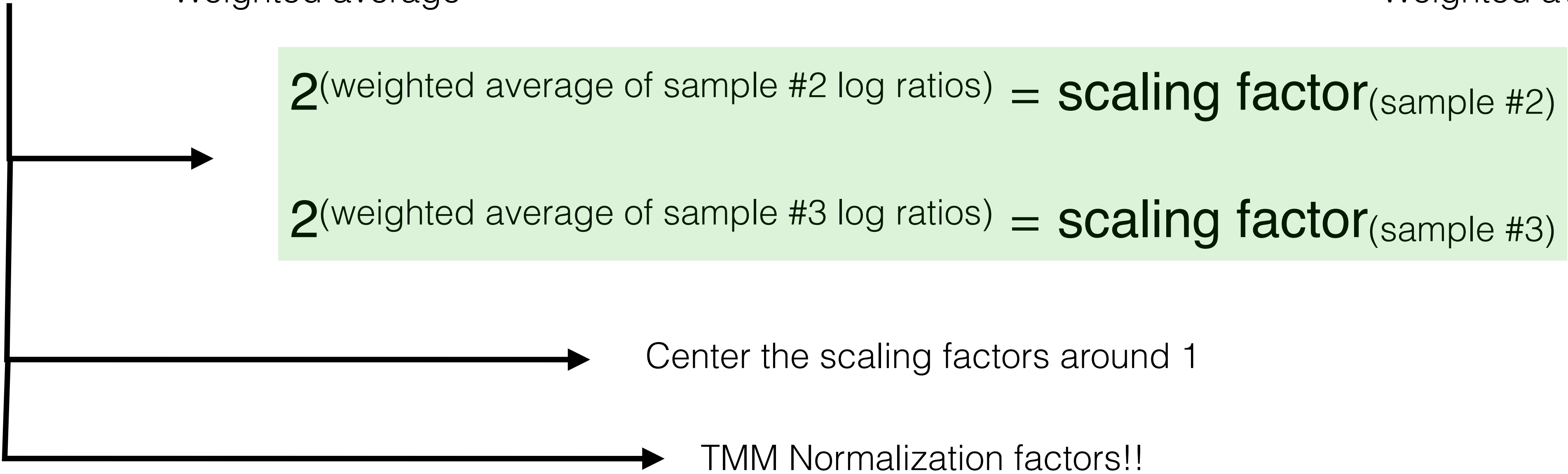
Step 4 - Calculate weighted average of the remaining log2 ratios

	$\log_2(\text{Reference} / \text{Sample \#2})$
Gene A	#
Gene B	#
Gene C	#
...	...

Weighted average

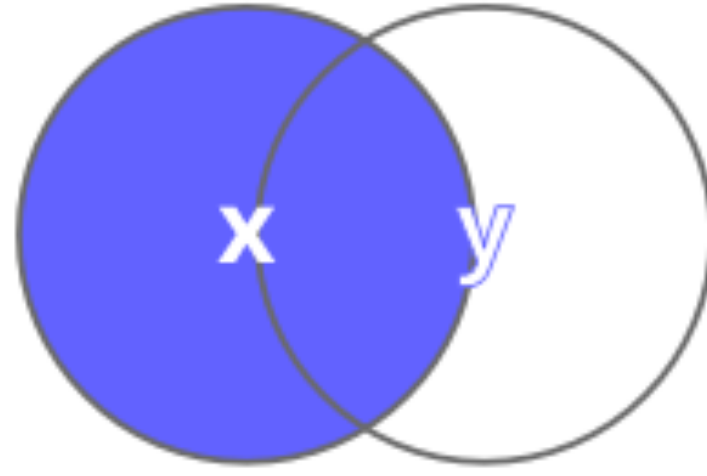
	$\log_2(\text{Reference} / \text{Sample \#3})$
Gene A	#
Gene B	#
Gene C	#
...	...

Weighted average

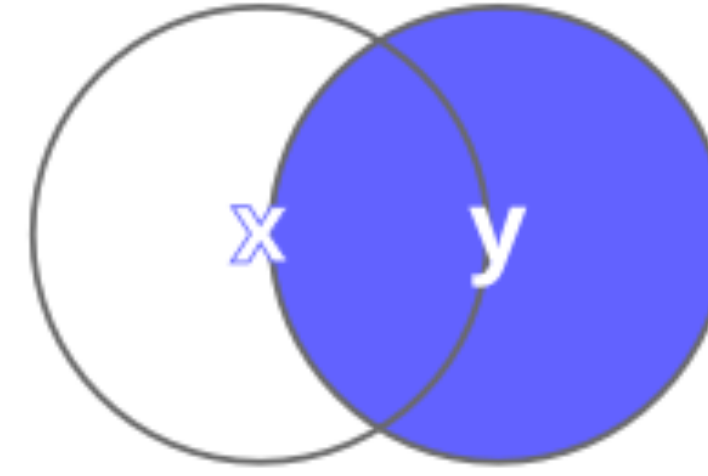


dplyr *joins*

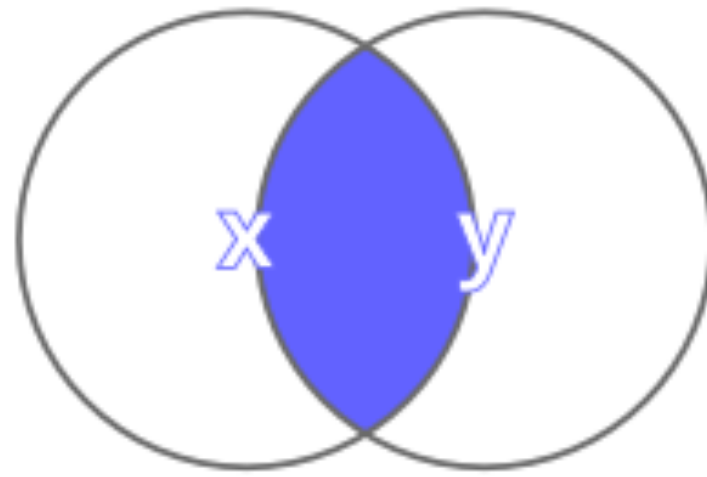
left_join(x, y)



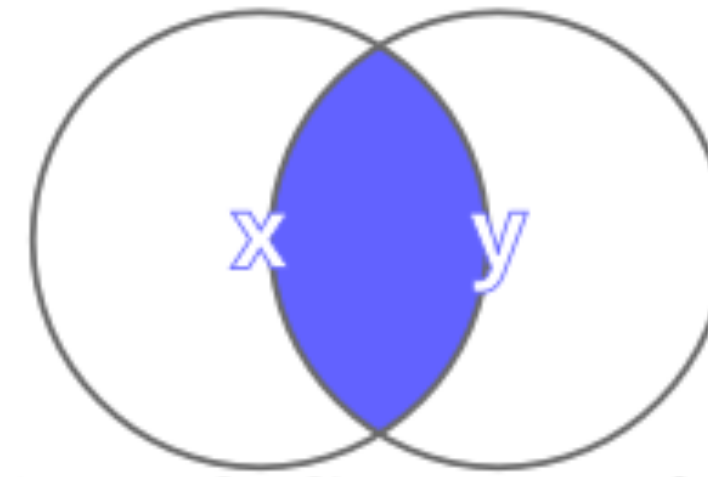
right_join(x, y)



inner_join(x, y)

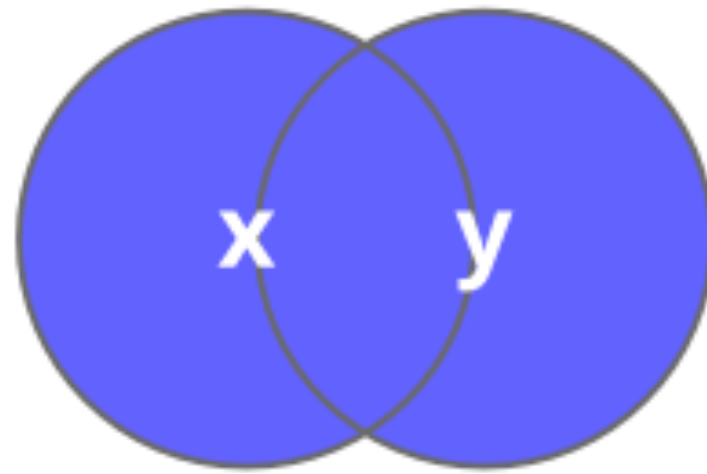


semi_join(x, y)



(never duplicate rows of x)

full_join(x, y)



anti_join(x, y)

